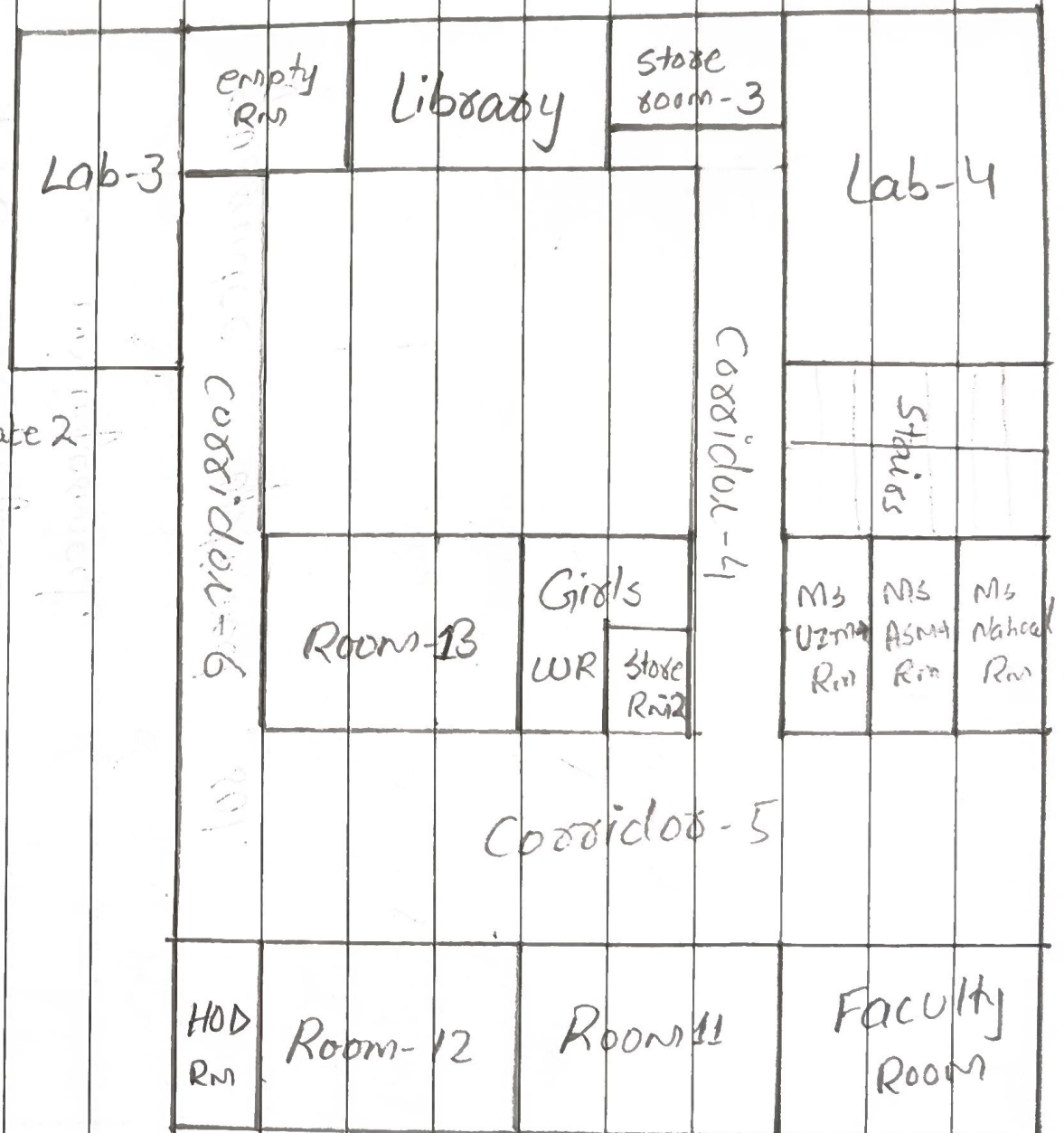


Date: .....

# Ground Floor



# First Floor



Problem State:

Initial Point:

HOD Room

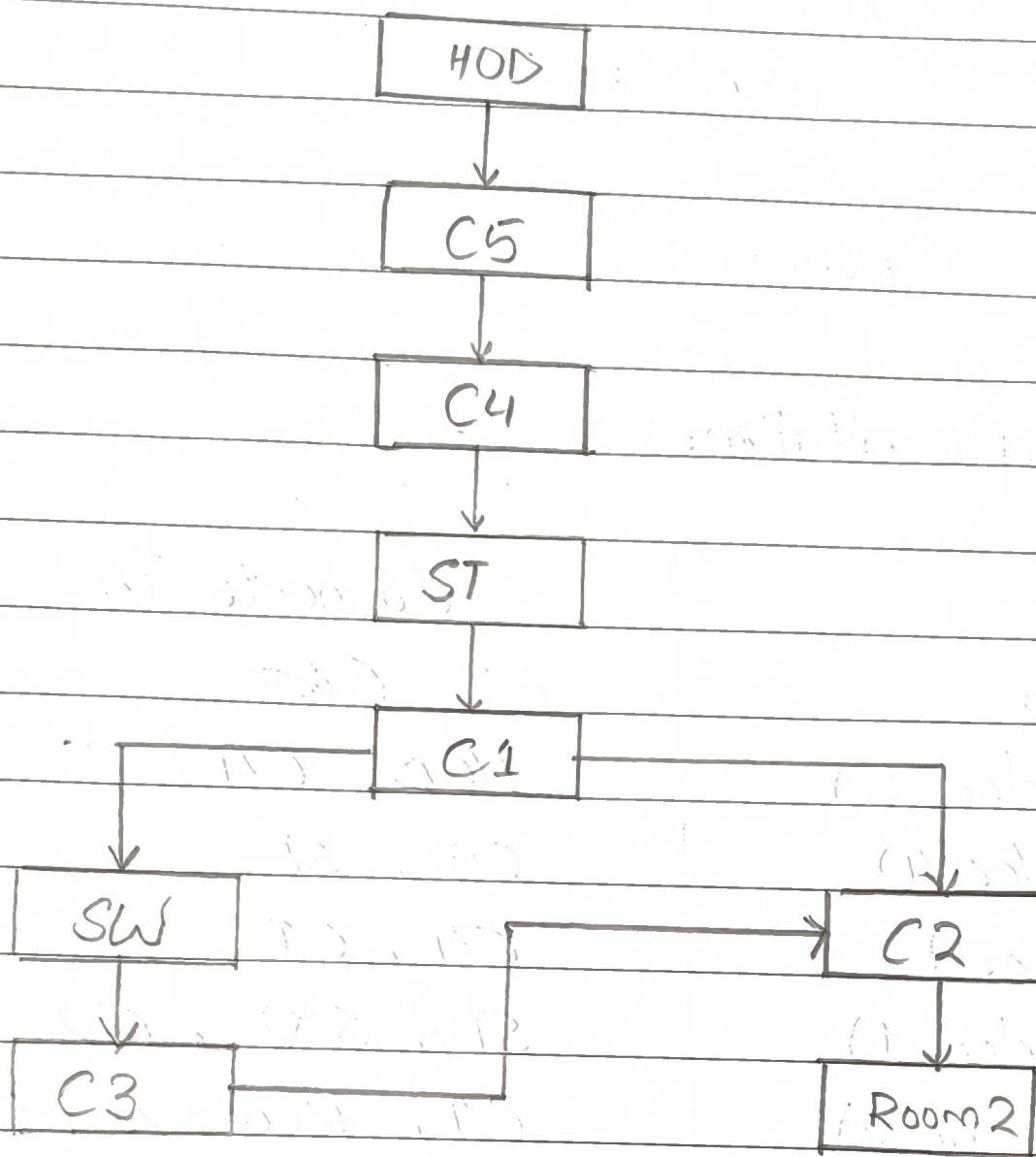
Goal :

Room - 2

Graph Representation:

| Node                          | Connects To                |
|-------------------------------|----------------------------|
| HOD Rm                        | C5                         |
| C5 (corridor-5)               | HOD, C4                    |
| C4 (corridor-4)               | C5, ST                     |
| ST (stairs)                   | C4, C1                     |
| C1 (corridor-1)               | ST, <del>SW</del> , C2, SW |
| C2 (corridor-2)               | C1, Room 2, C3             |
| SW (side walk)                | C1, C3                     |
| <del>C3</del> C3 (corridor 3) | <del>C3</del> SW, C2       |
| Room 2                        | C2                         |

BFS Tree:



BFS Trace:

Step 0: Initialize

 $Q = [HOD]$  $V = [HOD]$ S1: Deq HOD  $R_m$  expand  $HOD \xrightarrow{R_m} (C5)$  $Q = [C5]$  $V = [HOD, C5]$

S2 = Deq C5 expand C5  $\rightarrow$  (HOD, C4)

HOD Room already visited

Q = [C4]

V = [HOD, C5, C4]

S3 = Deq C4 expand C4  $\rightarrow$  (C5, ST)

C5 already visited

Q = [ST]

V = [HOD, C5, C4, ST]

S4 = Deq ST expand ST  $\rightarrow$  (C4, C1)

C4 already visited

Q = [C1]

V = [HOD, ---, ST, C1]

S5 = Deq C1 expand C1  $\rightarrow$  (S2, SW, ST)

ST already visited

Q = [C2, SW]

V = [HOD, ---, C1, C2, <sup>SW</sup>]

S6 = Deq C2 expand C2  $\rightarrow$  (C1, Room2, C3)

C1 already visited

Q = [SW, Room2, C3]

V = [HOD, ---, C1, C2, SW, Room2, C3]

S7 = Deq SW expand SW  $\rightarrow (C1, C3)$

Both are already visited

Q = [Room 2, C3]

V = [HOD, \_\_\_\_\_, Rm2, C3]

S8 = Deq Room 2 expand Room 2  $\rightarrow (C2)$

C2 already visited

S9 = Deq [C3] expand C3  $\rightarrow (C2, SW)$

both are visited

Goal achieved.

Shortest Path:

HOD  $\rightarrow$  C5  $\rightarrow$  C4  $\rightarrow$  GT  $\rightarrow$  C1  $\rightarrow$  C2  $\rightarrow$  Room 2

= 6 edges / steps

Alternate Path:

HOD  $\rightarrow$  C5  $\rightarrow$  C4  $\rightarrow$  GT  $\rightarrow$  C1  $\rightarrow$  SW  $\rightarrow$  C3  $\rightarrow$  C2

$\rightarrow$  Room 2

= 8 edges / steps.